

Pipeline Crossing Assessment Report

Assessment: Link check
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ASSESSMENT RESULT	PASS
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Calculated Outputs

Assessment stress at MAOP	25,903 psi
Assessment stress ratio	49.81% of SMYS
CSA Z662 class limit	80% SMYS (41,600 psi)
Live load hoop	1,474.9 psi
Hoop stress subtotal	25,903 psi
Bending strain stress	0 psi
Soil hoop	1,388.1 psi
Pressure hoop	23,040 psi
Fatigue status	Not checked
Fatigue damage ratio	0
Mitigated assessment	Not applied
Mitigation load factor	1

Pipeline Inputs

Outside diameter	24 in
Wall thickness	0.375 in
MAOP	720 psig
Class location	1
Design factor	0.8
SMYS	52,000 psi
Young's modulus	30,000,000 psi

Vehicle Inputs

Vehicle type	wheel
Number of axles/stations	2

Vehicle weight	23,000 lb
Contact area	180 in ²
Axle width	72 in
Crossing angle	90 deg
Impact factor	1.15

Soil, Strain, And Mitigation Inputs

Soil profile	compacted_bare
Soil load equation	prism
Depth of cover	48 in
Bending strain	0 microstrain
Mitigation type	none
Mitigation width	0 in
Mitigation length	0 in
Mitigation thickness	0 in

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